

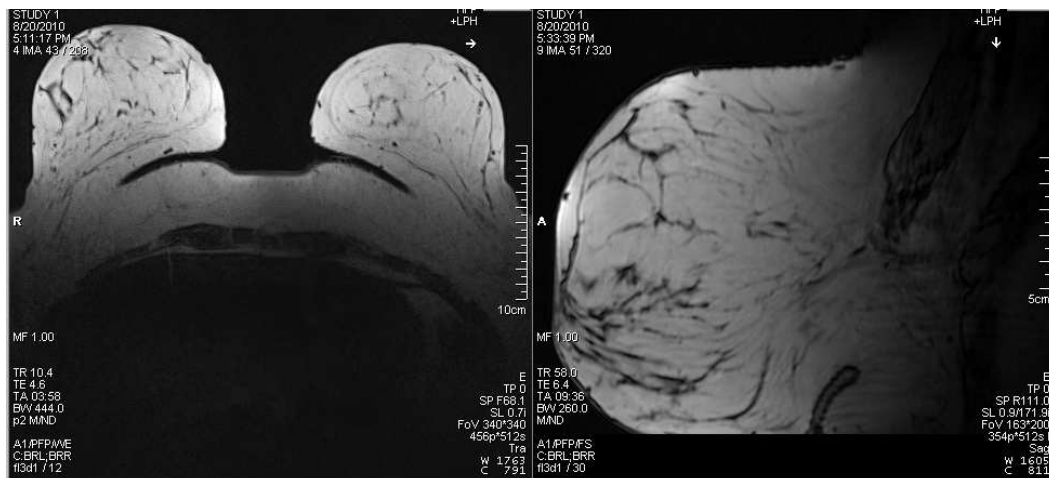
I. Problems with fat saturation

Fat saturation with spectral selective fat suppression might fail from time to time. We are dealing with several types of failing fat saturation:

- **local fat sat failure:** e. g. near coil elements or at skin folds or clips
- **regional fat sat failure:** e. g. one breast is saturated and the other breast not
- **global fat sat failure:** both breasts aren't saturated at all

i. *global failure of fat saturation*

In case of **global failure** of the fat saturation **the reason in most of the cases is an incorrect usage of the option "confirm frequency adjustment"** (error or misunderstanding) or it wasn't used at all. **There is no need to exchange the breast coil at first!**



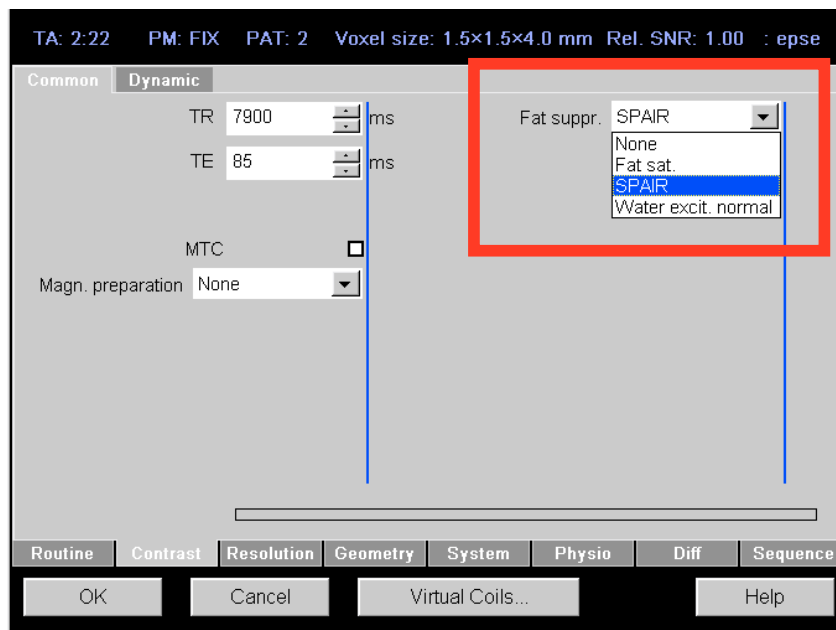
Global failing of spectral fat saturation

Please make sure that ideally at least two of the techs onsite are able to correctly apply and explain (to their colleagues) the "confirm frequency" popup-window. Show them the attached video 'frequency_adjustment_3t.avi' (be careful the frequency difference between fat and water for 1.5 T systems is 220 Hz and not 440 Hz as shown in the video). **In our experience the customer is calling at least two times until it is working. This is also valid for experienced user!**

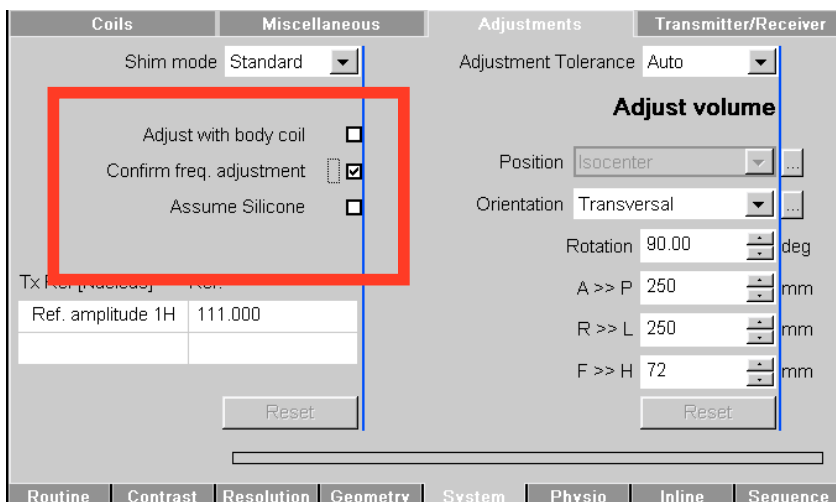
If the attached information and the **correct usage** of the option "confirm frequency adjustment" doesn't help you, please refer also to the attached **fat sat troubleshooting flowchart (attached fatsat_new.pdf)** for further assistance. By the way, this guide will also be very helpful in many other cases.

Application tips:

a) check all protocols



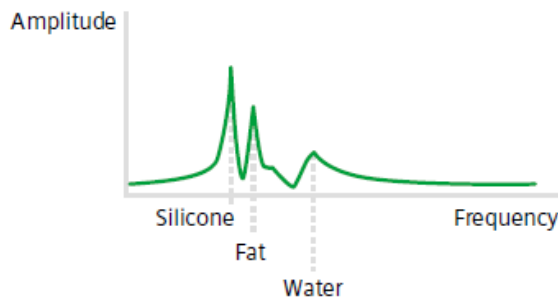
check for **each protocol** which uses **fatsat/spair/water excitation** (i.e. frequency selective fat suppression)



that in system/adjustments "confirm frequency adjustment" is checked: this has to be active for each of these protocols.

b) frequency adjustment

Usually, you should **see two peaks: fat (left) and water (right)**. If the patient has a silicone implantate, there is of course a third peak, which is located left of the fat peak:

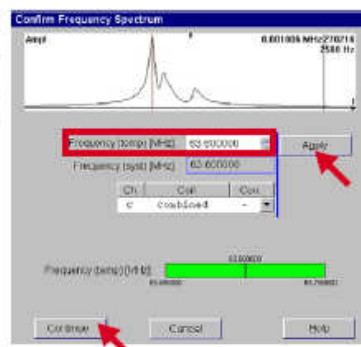


general display of spectral differences of water, silicone and fat

Display of MR spectra: The frequency of spectra (silicone, fat, and water) runs from left to right. Normally, the fat line somewhat overlays the silicone line.

AND

*Peaks from left
to right:
silicone—
fat—water*



how this is going to be displayed in the confirm frequency popup-window.

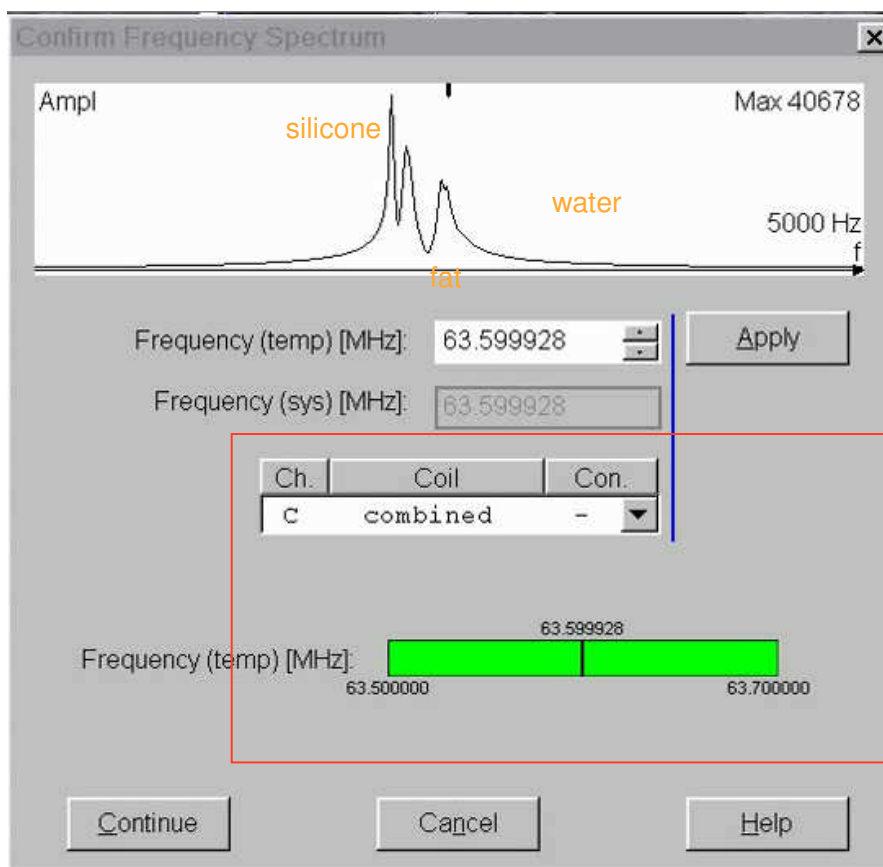
Click "Apply" and "Continue", to copy the desired frequency.

Depending on the age and breast tissue (fatty breasts), the detection of the water peak can fail, if the water peak is very small compared to the other peak(s): If the black tick mark (indicating the system frequency, within orange circle, see below) is located over the fat peak, the frequency has to be corrected.

If the wrong frequency was taken, there is most probably a huge difference between sequences with and without spectrally selective suppression (i. e. compare T1 vibs non-fat saturated and T2 tse fat saturated and the difference is about 200 Hz with 1.5 T).

c) Silicone Implants

- In case of silicone implants, 3 peaks should be identified (Silicone/Fat/Water)
- if system/adjustments/"assume silicone" is checked, the system tries to tune to water
- in (smaller) unilateral implants it may be necessary to select a coil element on the breast with the implant if the 3 peaks are not clearly identifiable using all elements combined
- there are techniques, where it is necessary to tune to the silicone frequency, e.g. using a watersat pulse to suppress silicone (SiliconeSat protocols). This is given in the tooltip of the protocols.

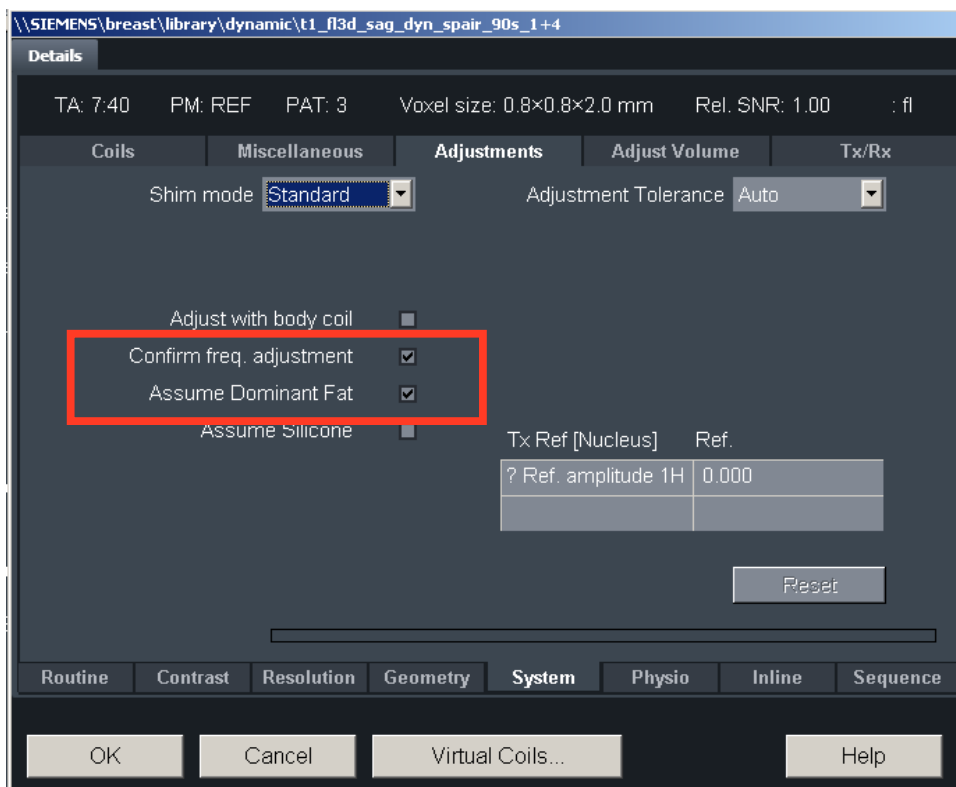


d) Improvements with VD11D

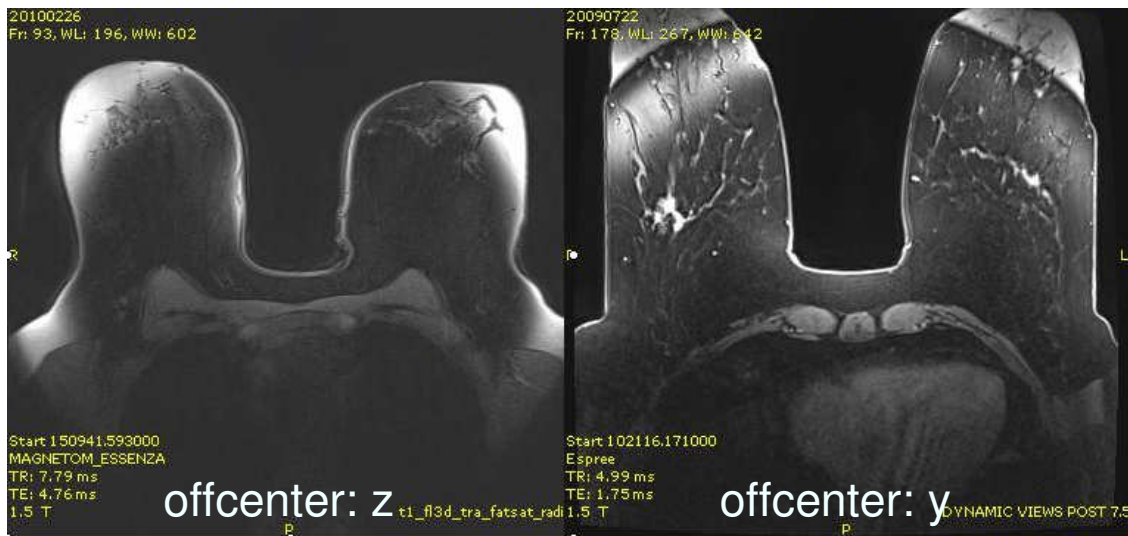
new system setting: assume dominant fat

- If only one peak is identified → this is automatically defined as fat peak, and 220/440 Hz are added to system frequency (→ system is tuned to water)
- If two peaks are identified, the system is tuned to water as well.
- Assume dominant fat and assume silicone are mutually exclusive.

When customer protocols are imported, this setting is usually automatically activated for breast protocols. Nevertheless, make sure that customers are aware of this setting and make use of it.



e) other reasons for global failure of fat saturation



Artifact appearance if the patient position is too far from the isocenter

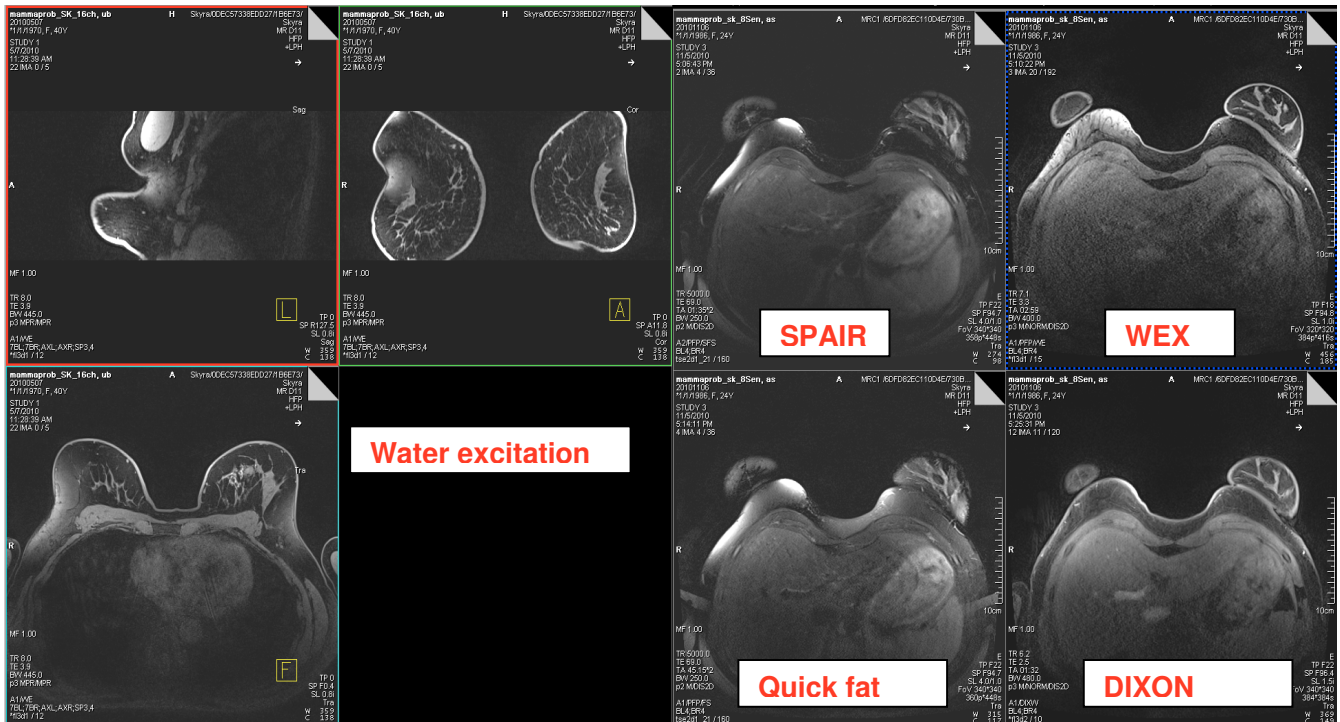
- a) in z-direction (FH)
- b) in y-direction (AP): here the breasts are touching the patient table (Espree riser was not used)

ii. Local failure of fat saturation

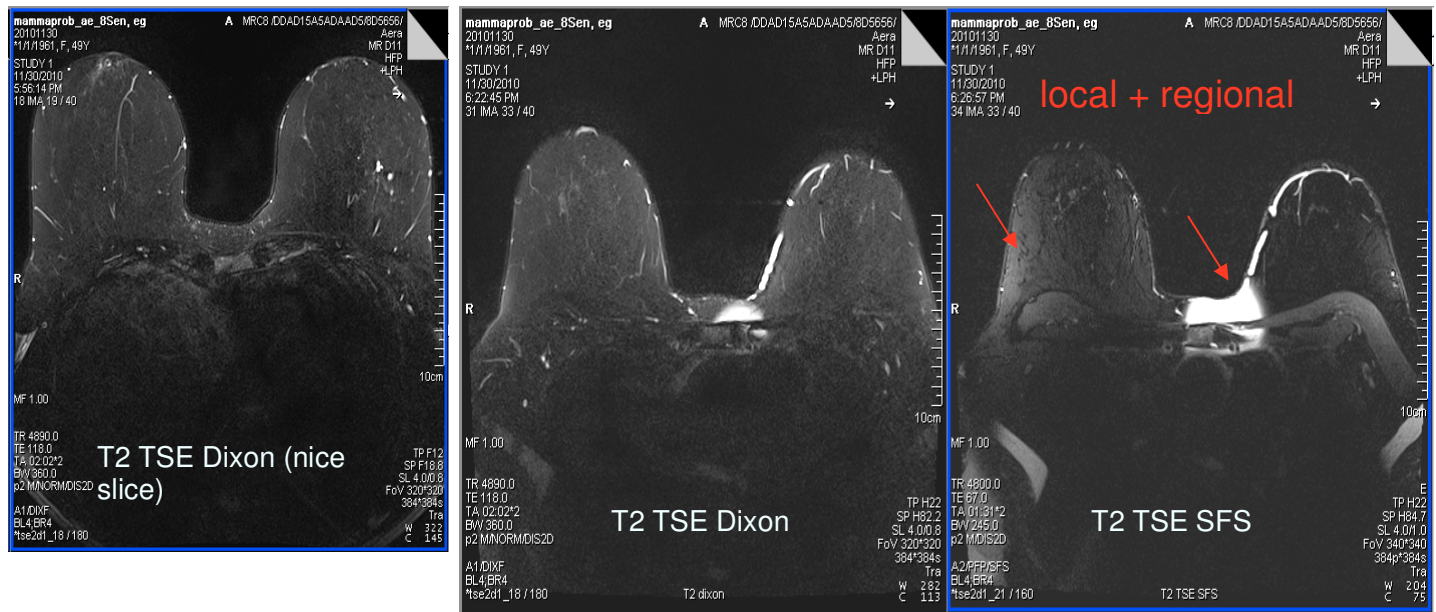
If you have local failure of fat saturation, please check if this is

- a) a region very close to a coil element
- b) a skin fold: Advise the customer to avoid skin fold as good as possible. In many cases it can be seen very clear after loading the affected series into the 3D-card and take a look at the MPR:
- c) metal clip

examples for skin folds



Example for metal clip:



Dixon T2-TSE can be a very good alternative in difficult B0 cases (here: metal clip) instead of SPAIR

iii. Regional failure of fat saturation

If the fat saturation fails in a specific region or at a specific side of the breast, please let the CSE check

- the date of the last retuning of ECC and CTC
- problems with the shield temperature within the last weeks or months?
- helium refill?

For further troubleshooting we would need also the following information:

- how many patients are affected, does it happen sporadically? When did it happen the first time?
- Are also problems known with other (fat saturated) exams in other regions?
- Any other problems like ghosting etc.?

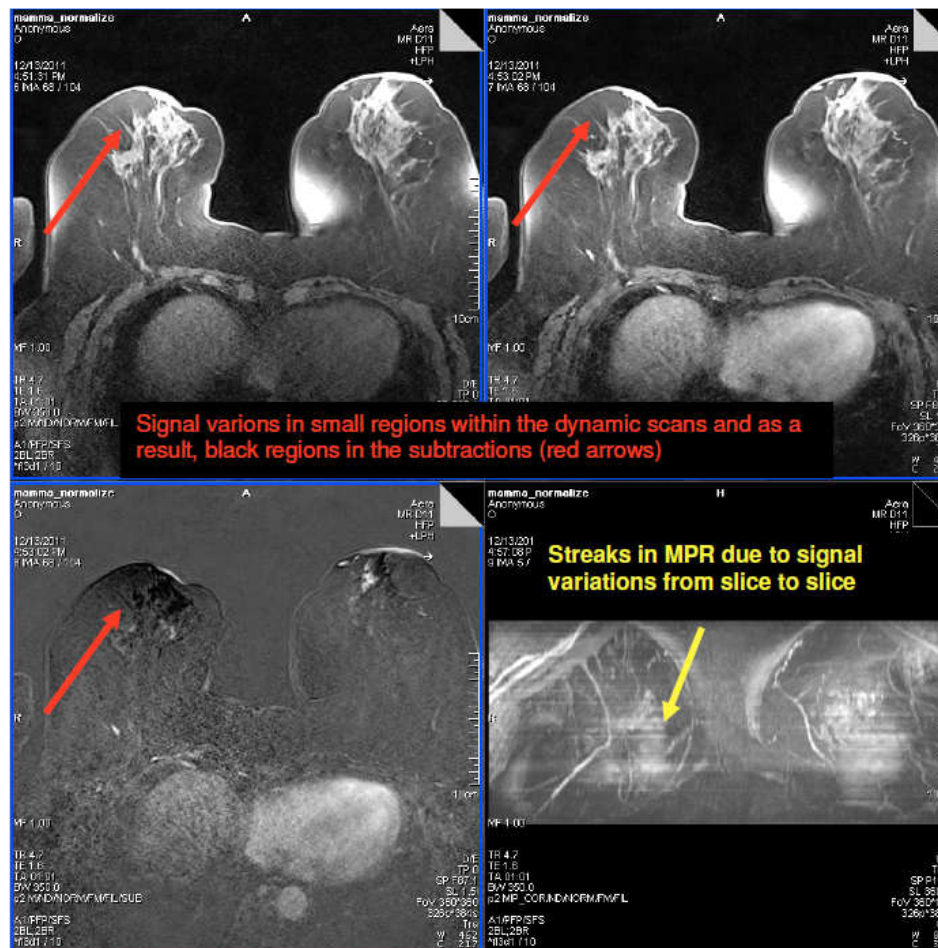
Please try also to make some tests (very easy changes in protocols like change of phase encoding direction etc.) from the **troubleshooting guide fatsat_new.pdf**

II. Normalization with Breast

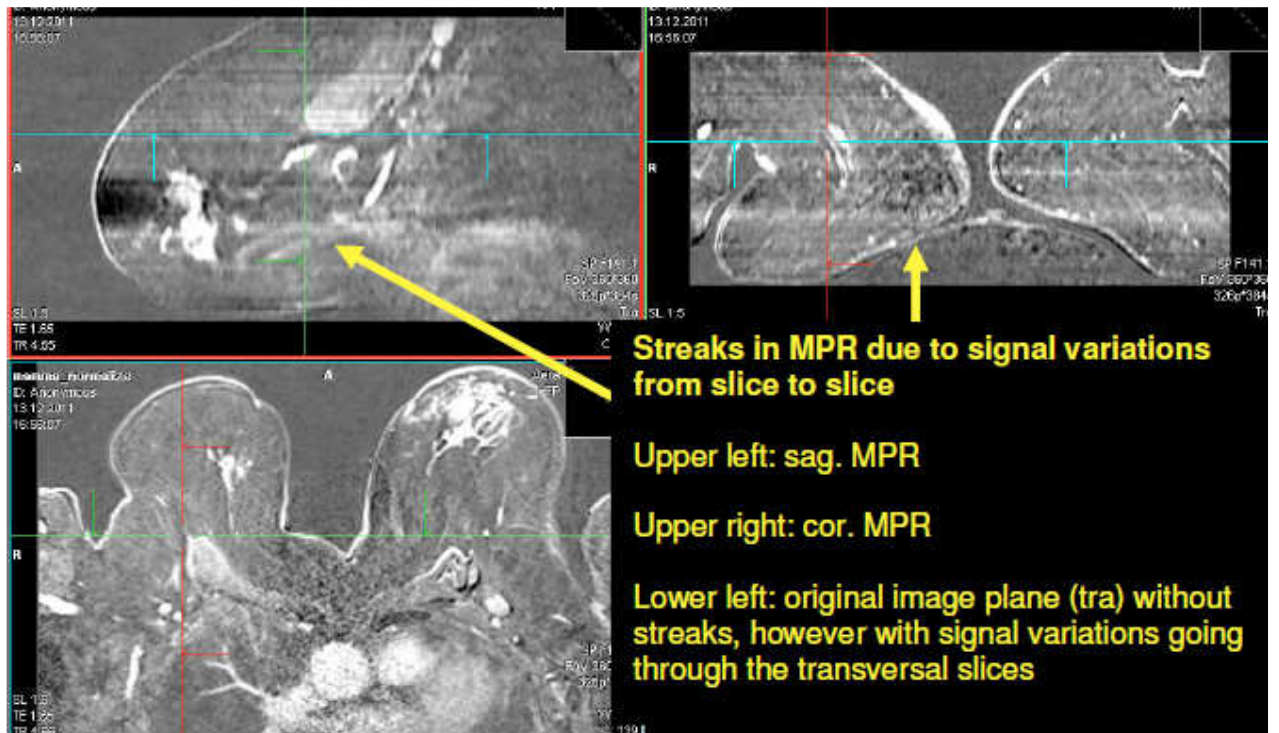
When accidentally selecting the **“Normalize” filter option instead of Prescan Normalize**, the resulting bad image quality are most of the times misinterpreted as a failing fat saturation. The following images will show you some examples.

The signal voids in the subtraction are caused by the fact, that the signal in this particular region is brighter in the pre-contrast than in the post-contrast scan.

The subtraction is calculated pixel-wise by the signal of the post-contrast minus the signal of a pre-contrast scan. If now the signal in the pre-contrast scan is higher than in the post-contrast measurement, the resulting pixel values are negative. **However, negative values cannot be displayed and will be set to zero --> which means "black" for the corresponding value in the pixel from the**



Ima1: Customer image, Aera, fl3d dynamic breast study. Only small signal variations comparing the native scan (upper left) with the postcontrast dynamic series (upper right). However, still signal voids as a result in the subtraction.



Ima3: Signal voids in the MPR (Upper left segment) including the streaks caused by the image-wise normalization due to the selected "Normalize" option. With Prescan Normalize, the normalization will be done over the whole volume.